De La Salle University - Dasmariñas
College of Information and Computer Studies

COURSE SYLLABUS
Second Semester SY 2024-2025

COURSE CODE:	**S-ITPE002**
COURSE TITLE:	**IT Professional Elective 2**
COURSE TYPE:	**Lecture**
COURSE CREDIT:	**2 units**
LEARNING MODALIT:	**Traditional Blended Learning**
PRE-REQUISITES:	**None**
CO-REQUISITES:	**S-ITPE002LA**
PROFESSOR:	**Maria Patricia B. Lina**
CONSULTATION HOURS:	**pblina@dlsud.edu.ph;**
	Thursday 16:00 – 18:00 MSteam
	Monday 1300-1400 COS100c

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course introduces the basic concepts and core principles used to formation of modern computer operating systems. The course aims to cover the skills and explain those concepts and principles that are prospective to endure for future years. , The course introduces the concepts of processes, concurrency, synchronization, scheduling, memory management, and file systems.

COURSE LEARNING OUTCOMES:

By the end of the semester, the students can:
CLO1 Identify the system concept, organization, operation. The Operating system evolution, security and process.
CLO2. Apply common scheduling algorithms and Access memory management
CLO3. Develop students' skills and knowledge in operating system implementation, table structure and device input output system
CLO4. Apply storage management, file, directories and solve storage management problem
CLO5. Developing student critical thinking through the creation of a design paper for a proposed server solution
CLO6. Apply the skills in implementing different kinds of OS, and its concept and properties.

LEARNING PLAN:

Course Learning Outcomes	Topic Learning Outcomes
CLO1 Identify the system concept, organization, operation. The Operating system evolution, security and process.	TLO1. Explain and describe the basic organization and abstract view of computer systems TLO2. Discuss the concepts in (operating) systems and OS major functions.
CLO2. Apply common scheduling algorithms and Access memory management	TLO3. Discuss the evolution of operating system, its protection to users, to program, and its standard security. TLO4. Introduce CPU scheduling, the basis for multi-programmed operating system and the CPU I/O burst cycle. TLO5. Explain pre-emptive and non-preemptive scheduling plus some different metrics used to make scheduling Decisions.
CLO3. Develop students' skills and knowledge in operating system implementation, table structure and device input output system	TLO6. Discuss how to apply several common scheduling algorithms TLO7. Explain the use of measurement and prediction for unknown scheduling parameters TLO8. Explain hardware required for memory protection and the concepts of logical and physical addresses
CLO4. Apply storage management, file, directories and solve storage management problem	TLO9. Discuss the problem of address binding, concept of segmentation and fragmentation. TLO10. Explain what paging is, its implementation
CLO5. Developing student critical thinking through the creation of a design paper for a proposed server solution	TLO11. Discuss different kinds of table structured and paging interact with segmentation. TLO12. Described benefits of virtual memory system. TLO13. Explain the concepts of demand paging, the working set model and memory replacement and allocation.
CLO6. Apply the skills in in implementing different kinds of OS, and its concept and properties	TLO14. Explain the general structure of the I/O subsystem and the different ways of performing I/O including polling, interruptions, and direct memory access. TLO15. Explain the nature of mass storage, disk storage management, concepts of files, directories and directories.

Midterm Period:

Module No and Learning Outcome(s)	Week Date/s	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Onsite	Asynchronous	Allotted Time
Module 0,1 CLO 1 CLO 6 TLO 1 TLO 2 TLO 3	Week 1 Aug 19 - Aug 24	• Gospel Reading, Reflection and Prayer • Module 0 - Introduction to the course • Module 1 Introduction to Operating System • System Organization • System Operation	MS Teams PowerPoint	/		2
Module 1 ,2 CLO1 CLO 6 TLO1 TLO 2 TLO 3	Week 2 Aug 26 - Aug 31	• Gospel Reading, Reflection and Prayer Module 1 • System concepts • What is an Operating System? Module 2 Structure and Protection • OS evolution • Kernels • Security Class Participation: Explanation of System Operation	MS Teams Schoolbook PowerPoint		/	2
Module 3 CLO 1 CLO 2 CLO 6 TLO 1 TLO3	Week 3 Sept 2 - Sept 7	• Gospel Reading, Reflection and Prayer • Module 3 – Processes • What is a process? • Process lifecycle • Inter-Process Communication (IPC) Class Participation: Explanation Operating System Process	MS Teams Schoolbook PowerPoint		/	2
	Week 4 Sept 9 - Sept 14	Enabling Assessment 1 Module 1,2,3 Quiz: Multiple choice and fill in the blank. 1 Hour Quiz	MS Teams Schoolbook PowerPoint	/		2
Module 4 CLO 2 CLO 3 CLO 5 TLO 4 TLO 5 TLO 6	Week 5 Sept 16 - Sept 21	• Gospel Reading, Reflection and Prayer Module 4 – Scheduling • Queues • Scheduling	MS Teams Schoolbook PowerPoint		/	2

TLO 7		• Multiple processor scheduling				
Module 5 CLO 2 CLO 3 CLO 5 CLO 6 TLO 4 TLO 5 TLO 6 TLO 7	Week 6 Sept 23 - Sept 28	• Gospel Reading, Reflection and Prayer Module 5 – CPU Scheduling Algorithms Explain the following: • First-Come First-Served (FCFS) • Shortest Job First (SJF) • Shortest Remaining Time First (SRTF) Class Participation: Compute for the FCFS and SJF	• MS • Teams • Schoolbook PowerPoint, visual expert	/		2
Module 5A CLO 2 CLO 3 CLO 5 TLO 4 TLO 5 TLO 6 TLO 7	Week 7 Sept 30 - Oct 5	Gospel Reading, Reflection and Prayer Module 5A – – CPU Scheduling Algorithms Explain the following: • Round Robin (RR) • Priority scheduling • Multilevel queues Class Participation: Design Round Robin Algorithms	MS Teams Schoolbook PowerPoint		/	2
CLO 2 CLO 3 CLO 5 TLO 4 TLO 5 TLO 6 TLO 7	Week 8 Oct 7 - Oct 12	Gospel Reading, Reflection and Prayer Midterm Enabling Quiz 2 Module 4,5,5A OS Scheduling computing Fill in the Blank, Multiple Choice and matching type.	MS Teams Schoolbook PowerPoint	/		2
Modules 1-5	Week 9 Oct 14 - Oct 19	Summative assessment: Module 1,2,3, 4,5,5A Fill in the Blank, Multiple Choice and matching type	s Schoolbook	/		2
			TOTAL	10	8	18

Final Period:

Course Learning Outcomes	Topic Learning Outcomes
CLO3. Develop students' skills and knowledge in operating system implementation, table structure and the input output system and devices.	TLO9. Discuss the problem of address binding, concept of segmentation and fragmentation. TLO10. Explain what is paging, its implementation
CLO4. Apply storage management, file, directories and solve storage management problem	TLO11. Discuss different kinds of table structure and paging interact with segmentation. TLO12. Described benefits of virtual memory system.
CLO5. Developing student critical thinking through the creation of a design paper for a proposed server solution	TLO13. Explain the concepts of demand paging, the working set model and memory replacement and allocation. TLO14. Explain the general structure of the I/O subsystem and the different ways of performing I/O including polling, interruptions, and direct memory access.
CLO6. Apply the skills in in implementing different kinds of OS, and its concept and properties	TLO15. Explain the nature of mass storage, disk storage management, concepts of files, directories and directory

Learning Outcomes for Final Period

Module Title and Learning Outcome(s)	Week /Dates	Teaching-Learning Activities / Assessment Strategy	Technology Enabler	Onsite	Asynchronous	Allotted Hours
CLO 2 CLO 3 CLO 5 TLO 4 TLO 5 TLO 6 TLO 7	Week10 Oct 21 - Oct 26	Gospel Reading, Reflection and Prayer Module 6 – DeadLock • Deadlock Introduction • Deadlock Detection and Recovery • Deadlock, Starvation, and Livelock • Deadlock Prevention and Avoidance	MS Teams Schoolbook PowerPoint	/		2
INDEPENDENT LEARNING DAY Week 11 October 28- October 31						
CLO 3 CLO 5 TLO 8 TLO 9 TLO 10 TLO 11	Week 12 Nov 4 - Nov 9	Gospel Reading, Reflection and Prayer Module 7 – Memory Management • Memory protection • Memory allocation Class Participation: Case study in File management	MS Teams Schoolbook PowerPoint	/		2

CLO 3 CLO 5 CLO 6 TLO 8 TLO 9 TLO 10 TLO 11	Week 13 Nov 11 - Nov 16	Gospel Reading, Reflection and Prayer Module 8 – Memory Management Explain the following: • Paging • Non-contiguous allocation • Paging implementation Page table structure	MS Teams Schoolbook PowerPoint		/	2
Module 7 CLO 3 CLO 5 CLO 6 TLO 8 TLO 9 TLO 10 TLO 11 TLO 12	Week 14 Nov 18- Nov 23	Gospel Reading, Reflection and Prayer Module 9 - Virtual Memory • Paging • Non-contiguous allocation • Paging implementation • Page table structure Class Participation: Case Study in Paging Implementation	MS Teams Schoolbook PowerPoint	/		2
	Week 15 Apr 22- Apr 30	Gospel Reading, Reflection and Prayer Enabling Assessment Module 6,7,8 Objective Type Fill in the Blank, Multiple Choice and matching type	MS Teams Schoolbook PowerPoint	/		2
Module 8 CLO 3 CLO 4 CLO 5 CLO 6 TLO 12 TLO 13 TLO 14 TLO 15	Week 16 Dec 2- Dec 7	Gospel Reading, Reflection and Prayer Module 10 – Input / Output system and file Management Explain • I/O subsystem • I/O devices • Kernel data structures	MS Teams Schoolbook PowerPoint		/	2
Module 9 CLO 4 CLO 5 CLO 6 TLO 12 TLO 13 TLO 14 TLO 15	Week 17 Dec 9- Dec 11	Gospel Reading, Reflection and Prayer Module 11 – Storage and File management system • Disk management • Files • Directories	MS Teams Schoolbook PowerPoint		/	2
Modules 6-11	Week 18 Dec 12- Dec 18	Summative assessment Moduel : 6,7,8,9,10,11 Fill in the Blank, Multiple Choice and matching type	Schoolbook	/		2
			TOTAL	10	8	18

GRADING SYSTEM:

Midterm	**%**	**Final**	**%**

Enabling Assessments (Activities, Quizzes, Presentations, etc.)	50%	Enabling Assessments (Activities, Quizzes, Presentations, etc.)	50%
Summative Assessment	30%	Summative Assessment	30%
(Midterm Examination)		(Final Paper)	
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. **Wala pong** synchronous meeting.

REFERENCES:

Onsite References	
Call Number	Book Title
QA 76.76.O63 .Si32 2009	Silberschatz A, Galvin PB, Gagne G. 2018. Operating system concepts. Hoboken, Nj: Wiley.
QA 76.76.O63 .Si32a 2000	Silberschatz A, Galvin PB, Gagne G. 2003. Applied Operating System Concepts, Windows XP Update.
QA 76.76.O63 .Y165 2014	Yang L. 2018. The Art of Linux Kernel Design. CRC Press.
QA 76.76.O63 .N961o 1999	Nutt GJ. 1999. Operating System Projects Using Windows NT.

Online	GeeksforGeeks. 2024 Jun 12. Operating System Tutorial. GeeksforGeeks. [accessed 2024 Aug 1]. https://www.geeksforgeeks.org/operating-systems. Tutorialspoint. 2019. Operating System - Process Scheduling - Tutorialspoint. Tutorialspointcom. [accessed 2024 Aug 6]. https://www.tutorialspoint.com/operating_system/os_process_scheduling.htm GeeksforGeeks. 2021 Aug 18. Memory Management in Operating System. GeeksforGeeks. [accessed 2024 Aug 6]. https://www.geeksforgeeks.org/memory-management-in-operating-system

EBSCOHOST (Journals)	da G, de J. 2023. NTFS FILE SYSTEM: AN INTRODUCTORY ANALYSIS. Journal of Engineering Research. 3(15):2–6. doi:https://doi.org/10.22533/at.ed.3173152308052

Prepared: **Information Technology Department**
1st Semester 2024-2025

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